Quantification: 876_z76c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	9
interior lacuna count	8
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1818.83	1904.5	368.311	px^2
major_axis_length_px	lacuna shape	80.949	86.8233	11.0381	px
minor_axis_length_px	lacuna shape	29.7503	29.3475	3.9133	px
aspect_ratio	lacuna shape	2.7475	2.8097	0.4227	unitless
eccentricity	lacuna shape	0.926	0.934	0.0282	unitless
solidity	lacuna shape	0.8569	0.8645	0.0398	unitless
roots_count	headline	8.75	8	3.6547	count
ring_length_r30_px	headline	617	644.5	103.82	px
ring_length_r60_px	reported	1622.75	1697.5	336.304	px
owned_length_px	ownership-dependent	6074.69	6440.53	2433.89	px
edge_count	ownership-dependent	446.75	452	182.949	count
mean_edge_length_px	ownership-dependent	13.6891	13.5864	0.7516	px
perimeter_px	normalised	230.364	236.066	41.9093	px
ring_area_r30_px2	normalised	8291.25	8605	1296.25	px^2
ring_area_r60_px2	normalised	21764.9	22785	3029.55	px^2
in_frame_fraction_r30	normalised	0.9993	1	0.0019	unitless
in_frame_fraction_r60	normalised	0.9836	1	0.0463	unitless
ring_density_r30	normalised	0.0757478	0.076343	0.0147785	px^-1
ring_density_r60	normalised	0.0748978	0.0760475	0.0142703	px^-1
roots_per_100px_perimeter	normalised	3.7447	3.4312	1.1717	per 100 px
ring_attached_length_r30_px	network v2	589.375	613.5	106.244	px
ring_attached_length_r60_px	network v2	1446.75	1432	432.203	px
ring_length_w_r30_px	network v2	698.747	726.393	116.115	px
ring_length_w_r60_px	network v2	1824.62	1906.47	382.074	px
sholl_crossings_r10	network v2	14.5	14	2.9277	count
sholl_crossings_r20	network v2	18.375	19.5	3.9978	count
sholl_crossings_r30	network v2	20.75	22.5	5.5997	count
loop_count	loops, report only	61.375	65.5	28.2081	count
loop_edge_share	loops, report only	0.5166	0.5248	0.0945	share
ring_width_mean_r30_px	width	6.1508	6.1685	0.7195	px

Per field
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measure	value	units
analysed_area_px2	1.03268e+06	px^2
total_skeleton_length_px	68597	px
canalicular_length_density_per_px	0.066426	px^-1
junction_count	2347	count
junction_density_per_px2	0.00227272	px^-2
lacuna_count	9	count

lacunae_per_px2	8.72e-06	px^-2
skeleton_component_count	627	count
mean_component_length_px	109.405	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0745211	px^-1
field_density_without_flagged_per_px	0.068501	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	156	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.066426	px^-1
field_length_density_with_unattached_per_px	0.0665771	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	598	count
loop_edge_share	0.4378	share
edge_count	5187	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.179	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	56575	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3529	196.715	34.7861	5.655	0.9842	0.5697	-1.4629	16.8161	858.163	True
2	1841	89.0492	27.9031	3.1914	0.9496	0.8121	1.5494	146.123	381.449	False
3	2155	85.0968	33.783	2.5189	0.9178	0.8641	-1.5153	296.21	913.135	False
4	2003	88.5498	30.0304	2.9487	0.9407	0.8649	1.5644	297.345	325.757	False
5	1105	63.6459	23.8313	2.6707	0.9273	0.806	-1.5304	527.835	198.727	False
6	1847	70.5663	34.2892	2.058	0.874	0.8966	-1.5649	538.637	444.666	False
7	1962	88.7858	28.6646	3.0974	0.9464	0.8975	-1.4695	687.621	605.365	False
8								646.408	104.687	False
9								925.23	965.577	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	648	1549	4507.45	292	15.4365	3.688
2	16	666	1742	6461.98	427	15.1335	
3	9	451	1319	7881.67	562	14.0243	
4	11	771	1963	9262.26	701	13.2129	6.8527
5	7	679	1963	6914.17	542	12.7568	6.2961
6	10	677	1869	5859.68	429	13.6589	6.9063
7	7	623	1653	6419.08	475	13.5139	6.0409
8	5	553	1453	4750.83	364	13.0517	5.8388

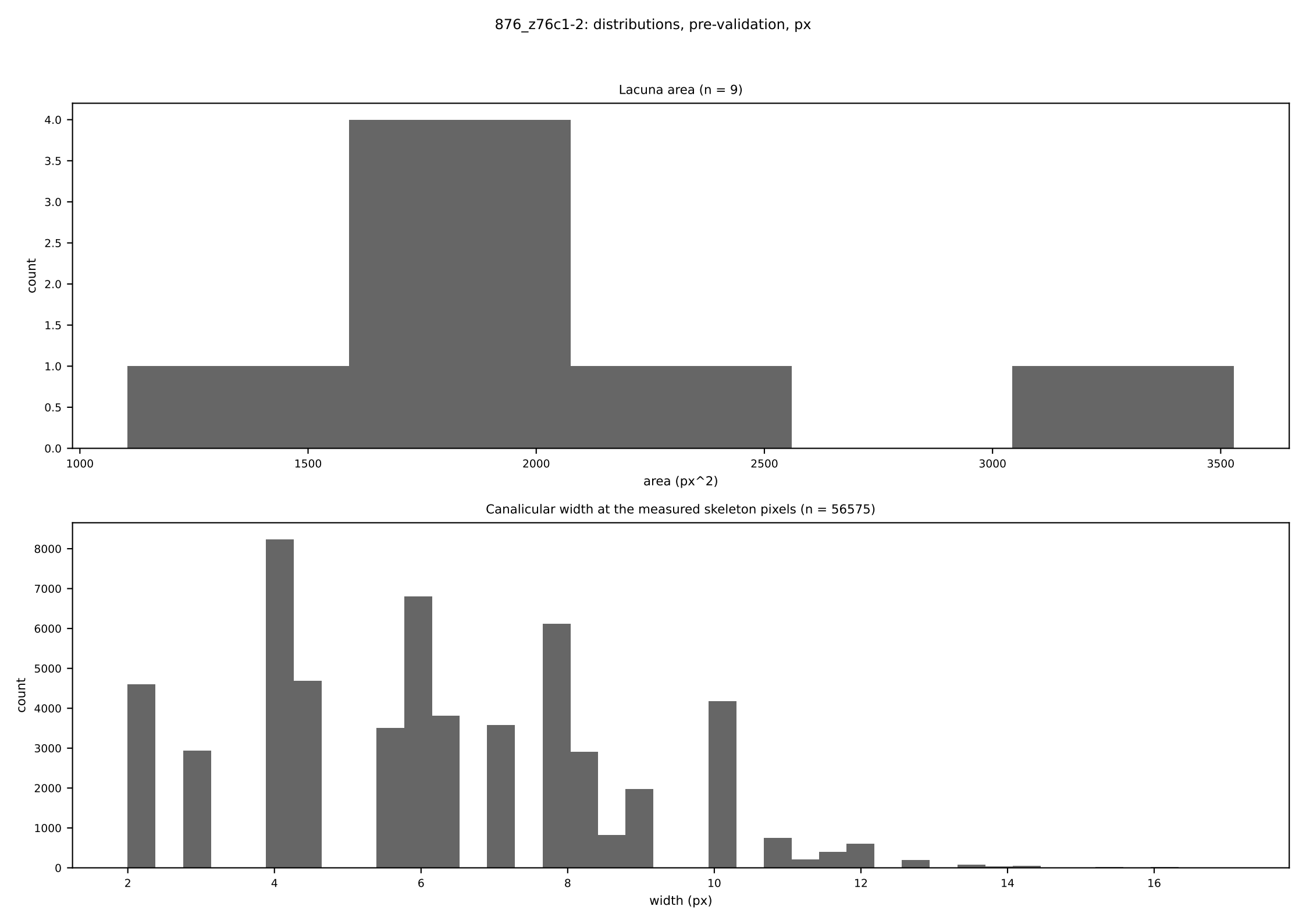
9 5 516 1020 1047.88 74 14.1605 4.9698

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	756.927	11162	23083	0.6571	0.6039	0.0580541	0.0671057	2.1138
2	269.948	9365	24218	1	1	0.0711159	0.07193	5.9271
3	274.883	9487	24500	1	1	0.0475387	0.0538367	3.2741
4	264.392	9452	24487	1	1	0.08157	0.080165	4.1605
5	195.072	7582	20596	1	1	0.0895542	0.0953098	3.5884
6	210.593	8239	22047	1	1	0.0821702	0.0847734	4.7485
7	249.12	8971	23523	1	1	0.069446	0.0702717	2.8099
8	155.894	5824	17026	1	1	0.0949519	0.0853401	3.2073
9	223.013	7410	17722	0.9946	0.8691	0.0696356	0.0575556	2.242

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	582	952	722.63	1720.24	17	16	19
2	637	1402	741.529	1933.41	18	21	21
3	402	915	497.99	1454.09	14	12	13
4	738	1843	872.37	2192.52	18	20	29
5	654	1957	765.458	2224.88	17	21	24
6	654	1833	760.872	2112.25	13	19	24
7	590	1462	711.257	1879.53	14	24	24
8	548	1388	645.843	1656.18	10	15	17
9	492	774	594.659	1144.13	12	15	14


876_z76c1-2: distributions, pre-validation, px
Lacuna area (n = 9)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 56575)
count
width (px)